

SDM College (Autonomous), Ujire-574240
Question Bank
Sub: Fundamentals of Computers
Class: I SEM BCA

UNIT I

2 Marks Questions:

1. Define Computer
2. Mention any four characteristics of Computers
3. What is meant by generations of Computers?
4. Expand UNIVAC, ENIAC.
5. What is diligence character of computer ?
6. What is a Microcomputer?
7. What is a Supercomputer?
8. What are Registers?
9. What is a Bus?
10. What is System Bus? Name its functional groups.
11. What is the main memory? Name two types.
12. Differential RAM and ROM.
13. What is Cache memory?
14. What are pointing devices? Give example.
15. What is Trackball? Why it is used?
16. What is Joystick? Mention its use.
17. What are output devices? List different forms of output devices.
18. What is a hard copy? What is a soft copy?
19. Write different purposes of the light pen.
20. What is a plotter? List different types of plotters.
21. Define resolution and refresh rate of monitors.
22. Describe any two differences between mainframe and super computers.
23. What are the components of a computer system?
24. What is CPU? What are its components?
25. How does a computer work?
26. What are the two main operations done by ALU?
27. Write a short note on keyboard.
28. Give differences between LCD and CRT.

5 Marks Questions:

1. List and explain characteristics of Computers.
2. List and explain in brief the characteristics of First-Generation Computers.
3. List and explain in brief the characteristics of Second-Generation Computers.
4. List and explain in brief the characteristics of Third-Generation Computers.

5. List and explain in brief the characteristics of Fourth-Generation Computers.
6. List and explain in brief the characteristics of Fifth-Generation Computers.
7. Write a note on (a) Mainframes (b) Supercomputers.
8. Draw the block diagram of a Computer System and explain the components of a Computer System.
9. List and explain the applications of Computers in different fields.
10. Write a note on a. Keyboard b. Pointing devices
11. Write a note on a. Touch Screen (b) Trackpad
12. Write a note on a. Digital Camera (b) Scanners
13. Write a note on a. Dot Matrix Printers (b) Ink-jet Printers
14. Write a note on a. Monitors (b) Laser Printer
15. Discuss the characteristic features of computers.
16. Explain the various types of computers in detail.
17. List out the various applications of computers and Explain.
18. Draw the block diagram of computer system and explain the function of various units of computer system.
19. What do you understand by the CPU? Describe in Detail the various units of the CPU.
20. What are registers in a CPU? Name five registers with their functions.
21. Explain communication between various units of a computer system.
22. What is system bus? Name the various units of the system bus.
23. What is Cache memory? Explain in detail.
24. Why are input and output devices necessary for a computer system?
25. What are the pointing devices? Discuss some of the commonly used pointing devices.
26. What is a touch screen device? Explain with their advantages and disadvantages and give some typical applications for which touch screen is most suitable as an input device.
27. What is speech recognition system? Explain its types.
28. What do you mean by optical scanning device? Explain any four such devices.
29. Explain the different types of printers. Make a clear difference between them in terms of speed, cost and method of operation, and suggest suitable application.
30. What is speech synthesizer? How does it function? Give some of its typical applications.
31. What is LCD? Differentiate between CRT and LCD monitors.

UNIT II

2 marks Questions.

1. Define System Software. Give example.
2. Define Application Software. Give example.
3. Define Utility Software? List any two System Utility functions.
4. Differentiate compilers and interpreters.
5. What is Machine Language?
6. What is Assembly Language?

7. What is Loader?
8. List basic tasks of a loader.
9. What is Word Processor? Give example for Word Processors
10. What is Spreadsheet Software? Give example for Spreadsheet Software
11. What are image editors? Give example .
12. What is Database Management System? Give example for Database Management System.
13. What is Presentation Software? Give example for Presentation Software.
14. What is Desktop Publishing? Give example for Desktop Publishing Software.
15. Define software? Mention the relationship between Software and Hardware.
16. Give categories of software and explain it.
17. Brief about third generation language.
18. What is Algorithm? Give example
19. What is Flowchart? Write the advantages of flowchart
20. What is problem analysis?
21. What is testing and debugging?
22. What is documentation and maintenance?
23. Explain the characteristics of algorithm.

5 Marks questions:

1. What do you understand by software? Discuss its types.
2. Explain the difference between hardware and software with their definition and uses.
3. What is operating system? Explain along with diagram.
4. What is Application software? Write a short note on application software.
5. Explain the program development cycle with the help of a block diagram.
6. Explain the characteristics of a good program.
7. Explain the feature of a good programming language.
8. Describe the classification of programming languages.
9. Write a short note on the following:
 - a) Compiler
 - b) Interpreter
 - c) Loader
 - d) Linker
10. What is third generation language? Give advantage and disadvantage high-level language.
11. Write a note on a. Device Drivers b. Language Translators.
12. List and explain in brief the advantages and limitations of Machine Languages.
13. List and explain in brief the advantages and limitations of Assembly Languages.
14. List and explain the advantages and limitations of High-level Languages.
15. Explain different 4GLs. Also, mention the advantages and disadvantages of 4GLs.

16. Write the steps involved in writing an algorithms and write an algorithm to find factorial of a number
17. Write an algorithm to check a number is prime or not
18. Write the advantages of flowchart and write different symbols used in flowchart and their meaning.
19. Explain the steps involved in problem solving process.
20. Write an algorithm to check for palindrome.
21. Explain Pseudo code with example. Write the advantages of Pseudo code

UNIT III

2 marks Questions.

1. What is number system?
2. Differentiate between 1's and 2's complement.
3. Convert $(1BC)_{16}$ to decimal.
4. Perform 10's complement subtraction on 24-09.
5. Using 2's complement, subtract $1000011-1010100$.
6. Convert $(BCA)_{16}$ to binary and octal.
7. Prove $x+x=x$.
8. Convert the following Octal Numbers to Binary and Hexadecimal:
a 154, 273, 256, 362, 267, 654, 745, 631, 537, 465 (2 Marks each)
9. Convert the following Hexadecimal Numbers to Binary and Octal: a. $(AB2)_{16}$, $(98D)_{16}$, $(AFD)_{16}$, $(1BC)_{16}$, $(AE5)_{16}$ (2 Marks each)
10. Convert the following Binary Numbers to Octal and Hexadecimal: a. $1010100(2)$, $101011(2)$, $1011110(2)$, $1111110(2)$, $110110010(2)$
11. Write 9's and 10's Complement of the following numbers a. 5243, 8126, 874, 8790, 4567.
12. Write 1's and 2's Complement of the following numbers a. $(1110101)_2$, $(101001)_2$, $(11001)_2$, $(111010)_2$, $(10101011)_2$.
13. Perform the following operations:
a. $(1101101)_2 - (1000011)_2$
b. $(101110111)_2 - (11011101)_2$

c. $(1101101)_2 - (11011)_2$

d. $(10011)_2 - (01001)_2$

e. $(100111)_2 - (100010)_2$ (2 Marks each)

14. State any two postulates of Boolean algebra

15. Prove that a. $X+X=X$ b. $X.X=X$ (2 Marks each)

16. Write the dual of a. $x + x = x$ b. $x + x' = 1$ (1 Mark each)

Descriptive questions:

1. Perform following subtraction using 9's and 10's complement method.

(i) $(8052)_{10} - (3250)_{10}$ (ii) $(6320)_{10} - (8659)_{10}$

2. Explain 2's complement subtraction with example.

3. State any 5 postulates of Boolean algebra.

4. Perform the following subtractions using 9's and 10's Complement Methods a. $841 - 329$
b. $325 - 641$ c. $821 - 413$ d. $8056 - 3250$ e. $6327 - 8659$ f. $6408 - 543$ g. $272 - 354$ (4 Marks each)

5. Perform the following subtractions using 1's and 2's Complement Methods (4 Marks each)

a. $(1001)_2 - (1011)_2$ b. $(10011)_2 - (1001)_2$ c. $(10110)_2 - (11101)_2$ d. $(110101)_2 - (100101)_2$

6. Perform following conversions. (4 Marks each) a. $(BCA.8)_{16} = ()_{10}$ b. $(201)_8 = ()_2$ c. $(451)_{10} = ()_8$ d. $(152.A1)_{16} = ()_{10}$ e. $(896.5)_{10} = ()_{16}$ f. $(345)_8 = ()_2$ g. $(915)_{10} = ()_8$ h. $(244.5)_{10} = ()_2$ i. $(111011010.11)_2 = ()_{10}$ j. $(625.5)_{10} = ()_{16}$

7. Using Venn diagram prove that a. $X=XY+X$ b. $x(y+z)=xy+xz$ c. $A+(B+C)=(A+B)+C$ (4 Marks each)

8. Prove state and prove commutative law using truth table.

9. State and prove distributive law. (AND) over + (OR) using truth table.

10. State and prove distributive law +(OR) over. (AND) using truth table.

11. State and prove DeMorgan's theorems for two variables.

12. State and prove DeMorgan's theorems for three variables.

13. State and prove any two theorems of Boolean algebra.

14. Write the procedure to perform 1's complement subtraction with example .
15. Write the procedure to perform 2's complement subtraction with example .
16. Write the procedure to perform 9's complement subtraction with example.
17. Write the procedure to perform 10's complement subtraction with example.
18. Using truth table prove first De Morgan's theorem $(x+y)' = x'y'$.
19. Demonstrate by means of truth tables the validity of the following theorems of Boolean algebra.
 - (a) The associative laws.
 - (b) The distributive law of + over.
20. Obtain the truth table of the function: $F = xy + xy' + y'z$.

UNIT IV

2 marks Questions

1. Write the Truth Table and Logic Symbol of the following AND gate, OR gate, NOT gate, NAND gate, NOR gate, XOR gate, XNOR gate (2 Marks each)
2. Draw the logic circuit for $F(X, Y, Z) = XY + X'Z'$ using basic gate.
3. Define Sum term and Product term.
4. What are minterms and maxterms?
5. What do you mean by the duality of the Boolean algebra? Give an example.
6. What do you mean by duality principle? Write the dual of $F = (x+y)(x+y')$
7. Write the general structure of 2 variables K-Map.
8. Write the general structure of 3 variables K-Map.
9. Write the general structure of 4 variables K-Map.
10. Define Pair, QUAD, and OCTET in K-Map.
11. Draw the logic circuit for the following Boolean functions using basic gates
 - a. $F(X, Y, Z) = XY + X'Z'$
 - b. $F(X, Y, Z) = (X+Y)'$
 - c. $F(A, B, C) = A'B + BC$
 - d. $F(X, Y, Z) = (X' + Y)Z$
 - e. $F(A, B, C) = AB' + A'C$ (2 Marks each)
12. Find the complement of Boolean expression $F = (X+Y)(X+Y'+Z)$
13. Find the complement of $F1 = x'yz' + x'y'z$
14. Find the complement of $F2 = x(y'z' + yz)$

15. Prove $x + x'y = x + y$
16. Simplify $xy + xy'$
17. What do you mean by standard form? Give an example.
18. What do you mean by canonical form? Give an example

Descriptive questions:

1. Express the Boolean function $F = XY' + X'Z$ as sum of minterms and product of maxterms.
2. Express the Boolean function $F = WXY' + X'Y'Z$ as sum of minterms and product of maxterms.
3. Using theorems and Postulates simplify $F(A, B, C, D) = A'B(D' + C'D) + B(A + A'CD)$
4. Using Boolean Theorems and postulates, prove the following: (5 marks each)
 - a. $x + x'y = x + y$
 - b. $x'y'z + x'yz + xy' = x'z + xy'$
5. Using truth table, prove the following: (4 marks each)
 - a. $x + x'y = x + y$
 - b. $x'y'z + x'yz + xy' = x'z + xy'$
6. Minimize the following Boolean expression using Boolean postulates and Theorems $F(A, B, C) = (A+B)(A+C)$
7. Implement Boolean function $F = x'y'z + x'yz + yz'$ with basic gates and also write the truth table.
8. Obtain the Truth Table of Boolean function and draw the logic diagram.
 - a. $F = xy + xy' + y'z$
 - b. $F = xy + x'y' + y'z$
9. Express the Boolean function, $F(X, Y, Z) = XY + X'Z$ as sum of minterms and product of maxterms.
10. Implement the following Boolean function using gates and also write the truth table.
 - a. $F = xyz'$
 - b. $F = xy' + x'z$ (5 marks each)

11. Express the Boolean function $F(A,B,C)=A+B'C$ as sum of minterms and product of maxterms.

12. Minimize the Boolean expression $F(x, y, z)=x'y'z + x'yz + xy'$ and draw the logic diagram for the minimized expression.

13. Solve the following using K-Maps and draw the logic diagram for minimized expression. Also draw the logic diagram for the simplified Boolean expression. (5marks each)

a. $F(A,B,C)=\sum(1,3,6,7)$

b. $F(A,B,C)=\sum(0,1,2,4,5,6)$

c. $F(A,B,C)=\sum 0,2,4,6$

d. $F(A,B,C)=\sum(1,2,5,7)+\sum(0,4,6)$

e. $F(A,B,C)=\sum(0,1,6,7)+\sum(3,4,5)$

f. $F(P,Q,R,S)=\sum(0,2,5,7,8,10,13,15)$

g. $F(A,B,C,D)=\sum(0,1,2,5,7,8,9,10,13,15)$
